



Precise measurements of the neutron magnetic form factor ¹

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Abstract

The neutron magnetic form factor G_{mn} has been determined via a measurement of the ratio of cross sections $D(e, e' n)$ and $D(e, e' p)$. The absolute detection efficiency of the neutron detector was measured with high accuracy using tagged neutrons produced from $H(n, p)n$ elastic scattering by means of a high intensity neutron beam. This approach minimizes the model dependence and improves upon the weakest points of previous experiments. Data in the range $q^2 = 0.2$ – 0.8 (GeV/c)² with uncertainties of $< 2\%$ are presented. © 1998 Elsevier Science B.V. All rights reserved.

1. Introduction

Precise data on the dependence of the nucleon form factors on the four momentum transfer q^2

provide detailed information on the inner structure of the nucleon and serve as a sensitive test for models of the nucleon. The proton form factors are known with excellent precision over a large range of momentum transfers. Due to the lack of a free neutron target, data for the neutron are of much poorer quality. This is true for both the electric form factor G_{en} and to a somewhat lesser degree for the magnetic form factor G_{mn} .

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G_{mn} has been determined mostly from quasi-elastic $D(e, e')$ cross sections [1–9]. The extraction of G_{mn} requires a longitudinal/transverse separation and a subtraction of the (dominant) proton magnetic contribution. The uncertainties resulting from the deuteron model, exchange currents (MEC), and final state interactions (FSI) are greatly amplified by the two subsequent subtractions. Mostly due to these theoretical uncertainties, the accuracy of G_{mn} is no better than $\sim 20\%$. An alternative approach, quasi-coincidence $D(e, e' \bar{p})$ [1,2,5], relies upon the *absence* of a recoiling proton $\bar{p}$ as identification of $(e, e'n)$ and requires an excellent understanding of the reaction mechanism, which is not at hand. More recently G_{mn} was extracted from a measurement of the asymmetry in ${}^3\text{He}(\vec{e}, e')$ [10] and from a cross section measurement of $D(e, e'n)$ [11]. The asymmetry experiment makes use of a new technique and the $D(e, e'n)$ experiment avoids the subtraction of the proton contribution. However, both measurements suffer from the limitations of past, the sensitivity to the structure of the nucleus employed as a neutron target.

To minimize the sensitivity to the nuclear structure one can extract G_{mn} from the ratio $R = (d\sigma(e, e'n)/d\Omega)/(d\sigma(e, e'p)/d\Omega)$ on the deuteron in quasi-free kinematics [12–15]. The $e-p$ cross section is well known and the ratio is insensitive to the wave function. Corrections due to FSI and MEC are small and well understood when restricting the acceptance to the quasi-elastic peak. The difficulty is shifted to the experiment: the precise determination of the *absolute* efficiency η of the neutron detector.

In other experiments the neutron detection efficiency has been determined employing the $D(\gamma, p)n$, $H(\gamma, \pi^+)n$, and $H(e, \pi^+)e'n$ reactions to produce a ‘beam’ of tagged neutrons [12–14,11]. However, it is not practical to determine $\eta(x, y, T_n)$ as a function of the neutron energy T_n , and the location of impact on the detector x, y because of the limitations in the intensity of these ‘beams’. In addition, for the case of $H(e, \pi^+)e'n$ used in [14], the 3-body final state introduced serious ambiguities [16]. Much of the complications in determining G_{mn} originates in the difficulties of such efficiency measurements. In fact, it has been argued that the recent data [14] suffer from a systematic error in the efficiency determination [16].

In our approach we employ a method [15] which improves upon the weakest point of the experiments: the determination of the neutron efficiency. Precise measurements of $\eta(x, y, T_n)$ and a detailed study of the detector response are possible when using high intensity neutron beams available at the proton-beam facilities. This has been demonstrated in a pilot experiment [15] where G_{mn} was determined with an accuracy of 1.7%.

2. Measurements at PSI

In the present work, the determination of the neutron detection efficiency was performed using a beam of 100–590 MeV neutrons available at the Paul Scherrer Institut (PSI) [17]. Neutrons are produced via the $C(p, n)$ reaction with a neutron flux (integrated over the entire energy range) of $10^8/s$ for a 10 μA proton beam and a 9 mm beam collimator. As in [15] the *tagged* high intensity neutron beam was produced via the $H(n, p)n$ reaction, scattering the neutrons from a 1 cm thick liquid H_2 -target. The recoil protons were detected with ΔE and E plastic scintillators recording the amplitude and the time-of-flight (TOF). Four multi-wire proportional chambers (MWPC) determined the target coordinates with an accuracy of ± 1.5 mm and the recoil angles with an accuracy of $\pm 0.2^\circ$.

The energy spectrum of the incident neutron beam was very different from the one used in [15] which had important consequences. The advantage of a wide energy spectrum was a simultaneous efficiency measurement over a large range of energies. However, due to the RF-structure of the proton beam (50 MHz) and the flight distance of 24 m, the determination of the neutron energy via TOF relative to the RF was not unique. This ambiguity was removed by measuring as well the TOF from the lH_2 -target to the E plastic bars. The precise calibration of the TOF was achieved with γ -rays produced in the neutron production target, converted to charged particles in a thin Pb-target at the position of the lH_2 -target, and detected with the ΔE scintillator.

The knowledge of the incident and the recoil energy, as well as the recoil angles provided a redundant determination of the energy and position of the tagged neutron beam which was free from

contributions of background reactions. The accuracy of the tagged energy was 2.1–15.6 MeV for the energy range (120–460 MeV) of the tagged neutrons and the position resolution on the detector surface was 7.0 mm. As in [15] the hadron detector was placed in this neutron beam and its efficiency measured. The detector consisted of two 10 cm thick plastic converters, E_f and E_b , preceded by 3 thin ΔE counters to identify the incident nucleon. Except for the opening towards the target, the detector was shielded with 10 cm thick lead walls. Lead absorbers of 1–20 mm thickness were placed at the entrance window of the detector. These absorbers are needed in the $D(e,e'n)$ measurements in order to absorb low energy photons. Use of the identical configuration at PSI ensured that absorption effects of neutrons were automatically included in the efficiency determined.

When using a neutron beam with a *continuous* energy spectrum, tagged scattered neutrons of the full range in energy are produced at all scattering angles. The high intensity of the tagged neutron flux allowed to study $\eta(x,y,T_n)$ as a function of x,y,T_n simultaneously, without moving the detector as was done in [15]. The determination of the absolute efficiency *distribution* $\eta(x,y,T_n)$ is necessary because the detector illumination is unavoidably different for the $H(n,p)n$ and the $D(e,e'n)$ data taking. An example of $\eta(x,y,T_n)$ is shown in Fig. 1.

The dominant source of the variation of $\eta(x,y,T_n)$ due to the light collection efficiency was measured

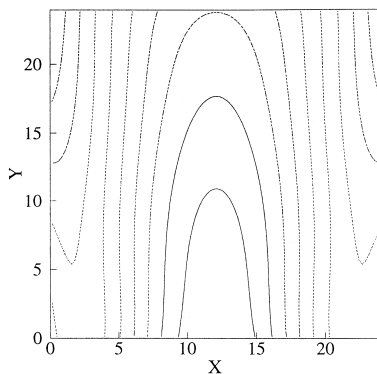


Fig. 1. Fit to the efficiency distribution $\eta(x,y,T_n)$ at $T_n = 270 \pm 30$ MeV. The efficiency changes by 1.5% between two lines of constant $\eta(x,y,T_n)$. The lightguide is attached at the top of the map.

separately using $H(n,p)$ recoil protons, as it needs to be known with good spatial resolution near the detector edges. This was done by placing the detector on the recoil arm and replacing the E scintillators by a second thin ΔE -detector to provide a tag for the recoil protons which does not depend on the neutron detector. The spatial resolution is provided by the four MWPC. The knowledge of $\eta(x,y,T_n)$ allows to match the efficiency η needed to determine R from the yield ratio with minimal corrections (see Table 1). If the efficiency η is measured only for a central illumination of the detector at the center value T_n of the $(e,e'n)$ -measurement this correction can be as large as 14%.

Tagged recoil protons were also used for several additional corrections which are needed in the determination of R over the entire energy range. In particular, multiple scattering and absorption effects of the protons, which are the dominant correction of R , were measured for all Pb-absorbers employed with an accuracy of 0.7% (Table 1). Whereas at low energy multiple scattering mostly reduced the $D(e,e'p)$ yield, absorption effects dominated in the present experiment due to the thicker lead absorbers and the higher proton energies employed. Table 1 summarizes all the measured corrections with its uncertainties.

3. Measurements at MAMI

The measurements of the $D(e,e'n)$ and the $D(e,e'p)$ yield in quasi-elastic kinematics for $q^2 = 0.235, 0.504, 0.652, \text{ and } 0.784 \text{ (GeV/c)}^2$ were performed at the Mainz Microtron (MAMI) [18]. As in [15] the $(e,e'n)$ and the $(e,e'p)$ yields were measured simultaneously which made the ratio independent of the luminosity, dead time effects, and the efficiency of the electron arm. The essential difference to [15] is the 100% duty factor of MAMI which is needed in order to achieve an acceptable signal-to-noise ratio for the $(e,e'n)$ measurement at higher q^2 . In the present experiment the signal-to-noise ratio for neutrons was better than 150, even at the highest q^2 .

The measurements were performed with a 0.1–1.2 μA cw electron beam incident on a cylindrical 2 cm thick liquid deuterium target cell with 6 μm HAVAR windows. The incident beam energy was

555 MeV for the lowest q^2 -point and 855 MeV for all higher q^2 -measurements. Spectrometer A [19] with a solid angle limited to 21 msr detected the scattered electron in coincidence with the recoiling hadron. The hadron detector covered an angular range of ± 78 mr horizontal and vertical resulting in a solid angle of 24.3 msr. The detector covered 50–90% of the Fermi cone depending on kinematics.

Considerable care was taken to obtain identical cuts on the energy deposited in the detectors. $H(e,e'p)$ at MAMI and $H(n,p)$ at PSI were used to obtain absolute calibrations of the amplitude spectrum. In addition, each converter was monitored for both gain variations and baseline shifts using two temperature compensated LED's whose light output was in turn monitored by very stable PIN-diodes [15]. The energy scale of the converters was identical to an accuracy of 0.6% in the PSI and the MAMI runs.

The off-line analysis of the $(e,e'n)$ and the $(e,e'p)$ yields were very similar to the analysis described in [15]. The threshold used in the definition for neutrons in E_f was set at 25 MeV. The number of counted neutrons was corrected for the efficiency of the veto condition and for misidentified protons due to inefficiencies of the veto counters. Depending on the kinematics the determined veto efficiency corrections are 1.6–10.8%. The correction for misidentified protons ranged from 0.8% to 4.9%. The three independent measurements for the number of neutrons determined with the three different combinations of ΔE counters agreed within 0.3%. The uncertainty in these corrections are small relative to the

statistical error of the number of neutrons and are included in the error of the yield ratio as shown in Table 1.

Protons were counted in the TOF spectrum of E_f with a coincident time signal in at least one of the three ΔE counters. The number of protons was corrected for the inefficiency of the ΔE counter used. The final number of protons, obtained using different $\Delta E, E_f$ combinations, agreed within 0.08%.

4. Evaluation of R

From the measured ratio of detected neutrons and protons, the ratio R of the $D(e,e'n)$ and $D(e,e'p)$ cross sections was determined, making use of the carefully determined absolute efficiency of the detector. Two efficiency measurements bracketed the measurement of the yield ratio at MAMI in order to check the reliability of η . Consistent results for η are found, and the resulting values of R are independent of the applied E_f threshold. Table 1 summarizes errors and applied corrections, all of which are based on measurements, and the final results for R .

To extract G_{mn} from the measured R values we have to go beyond the plane wave impulse approximation (PWIA) and take into account the effects of FSI, MEC, and isobar currents (IC). The results for R^{theo} of Arenhövel [20], obtained with the Paris potential, were used. The n-p final state interaction, mostly charge exchange scattering, gives the largest contribution of 1.3–4.0% to R depending on kinematics. The contributions from MEC and IC are of

Table 1

Results for R and the experimentally determined errors and corrections in %. St, Sy indicates whether the error is mainly statistical or systematical in nature

q^2 (GeV/c) ²		0.235	0.504	0.652	0.784
Error of the yield ratio	St	± 0.5	± 0.9	± 0.8	± 1.5
Illumination matching	Sy	± 0.4	± 0.6	± 0.7	± 0.8
Threshold calibration	Sy	± 0.8	± 0.8	± 0.8	± 0.8
Proton losses	Sy	-15.2 ± 0.7	-13.5 ± 0.7	-18.1 ± 0.8	-16.3 ± 0.8
(p,n)-correction	Sy	-0.14 ± 0.04	-1.1 ± 0.3	-2.1 ± 0.7	-2.1 ± 0.7
Error of η	St	± 0.86	± 0.98	± 0.82	± 0.51
T_n and x, y uncertainty	Sy	± 0.45	± 0.43	± 0.82	± 0.90
R		0.234	0.366	0.415	0.445
Relative error of R in %		± 1.5	± 2.0	± 2.0	± 2.3

Table 2

Results for G_{mn} relative to the dipole form factor $G_D = (1 + q^2/0.710)^{-2}$. The error on G_{mn} includes both the experimental contribution (table 1, σ_{e-p} , G_{en}) and the ones due to theory

q^2 (GeV/c) ²	0.235	0.504	0.652	0.784
$[(R_{\text{PWIA}}/R)^{\text{theo}} - 1](\%)$	-4.9 ± 0.6	-2.8 ± 0.4	-2.2 ± 0.3	-1.9 ± 0.2
Contribution of $G_{en}(\%)$	-2.0 ± 0.6	-2.4 ± 0.7	-1.8 ± 0.5	-1.2 ± 0.4
σ_{e-p}/σ_D	0.938 ± 0.009	0.958 ± 0.009	0.987 ± 0.008	1.022 ± 0.006
$G_{mn}/(\mu_n G_D)$	0.962 ± 0.009	1.032 ± 0.012	1.037 ± 0.012	1.043 ± 0.012

order 0.5% and relativistic effects are negligible. The total corrections are listed in Table 2 as a deviation $D = (R_{\text{PWIA}}/R)^{\text{theo}} - 1$ from the PWIA-value. The dependence of D on the nucleon–nucleon potential was studied at $q^2 = 0.235$ and 0.504 (GeV/c)² where the correction is largest. The variation found when using the Bonn-R-space, the Argonne V14, and the Nijmegen potential is well within the systematic uncertainties (FSI $\pm 15\%$, MEC $\pm 40\%$, IC $\pm 60\%$) used to calculate the errors of Table 2.

The resulting value $R_{\text{PWIA}} = (\sigma_{e-n}/\sigma_{e-p})_{\text{PWIA}} = R \cdot (1 + D)$ is the experimental ratio of the $e-n$

cross section (which is essentially given by G_{mn}) relative to the $e-p$ cross section corrected for non-PWIA contributions. The contribution of the neutron electric form factor to R is small; it introduces only a small additional uncertainty despite the poor knowledge of G_{en} (see Table 2).

To evaluate the $e-p$ cross section σ_{e-p} , we used the world's supply of σ_{e-p} -data in a range of 0.5 fm^{-1} around the desired q . In this range of q , we used a parameterization with a relative q -dependence as given by the Höhler fit [21] to G_{ep} and G_{mp} , with the overall normalizations fitted to the

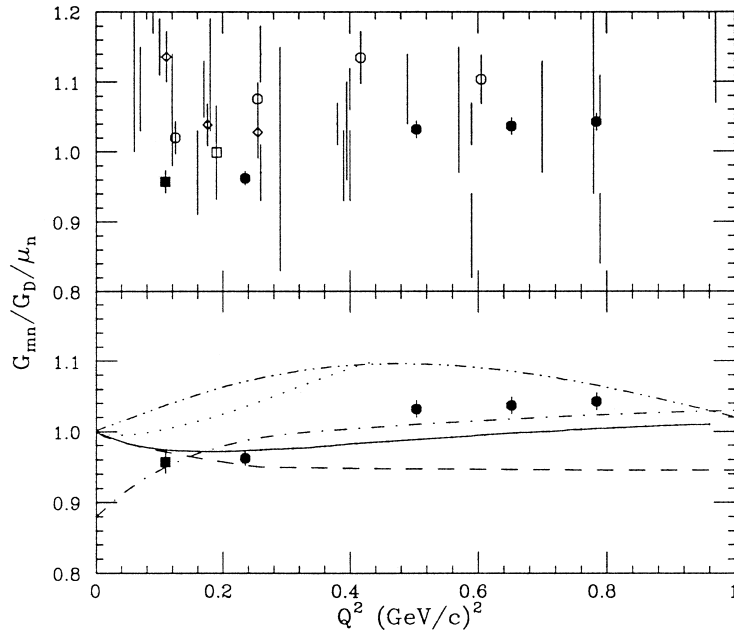


Fig. 2. Top figure shows the available data on G_{mn} relative to the Dipole parameterization. Shown are the data of [15] (solid squares), Gao et al. [10] (open squares), Markowitz et al. [11] (open diamonds), Bruins et al. [14] (open circles), and the present work (solid circles). Data measured before 1990 are only shown by the error bar. The bottom figure shows the data of Anklin et al. [15] (solid squares), and of the present work (solid circles), in comparison to various model calculations. Solid: Mergell et al. [22], dot: Meissner [23], dash-dotted: Eich [24], dash: Schlumpf [25], dash-dot-dotted: Lu et al. $r_{\text{Bag}} = 0.9 \text{ fm}$ [26].

world data. The statistical errors of the data have been treated in the standard way, while the systematic errors have been accounted for by changing the data of each set by its error, refitting, and adding the changes due to systematic errors in quadrature. The resulting $e-p$ cross sections are listed in Table 2 together with the final results for G_{mn} , which are plotted in Fig. 2. Both the proton and neutron form factors differ from the crude empirical expression $G_D = (1 + q^2/0.710)^{-2}$ used to remove the dominant q^2 -dependence in Table 2 and Fig. 2. The discrepancy of the present results of G_{mn} with data from other recent experiments [14,11] has already been discussed elsewhere [15,16].

The data of the present work and of Anklin et al. [15] show significant differences to both the non-relativistic constituent quark model calculation by Eich [24], and the relativistic version by Schlumpf [25]. Similar differences are observed when comparing the data to a recent cloudy bag model calculation by Lu et al. [26]. The data are also compared to the minimal vector dominance model by Meissner [23] and the recent calculation by Mergell et al. [22] based on a fit of the proton data using dispersion theoretical arguments. None of the calculations describe the data satisfactorily.

5. Conclusion

In the present experiment we determined G_{mn} in the q^2 -range from $0.23 (\text{GeV}/c)^2$ to $0.78 (\text{GeV}/c)^2$. We achieved a substantial decrease of the uncertainties. The improvement in the knowledge of G_{mn} is due to the fact that we use an approach that depends the least on the input of theory; this becomes possible by performing the needed calibrations using a high-intensity tagged neutron beam, and by using an electron beam with 100% duty factor. The results, listed in Table 2, indicate that accuracies of $< 2\%$ of G_{mn} can be reached. This increase of accuracy allows for a detailed analysis of the electromagnetic structure in terms of theoretical model calculations.

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